

G

LV.0 → LV.1 · AI ENGINEER

The 6 G's that move you from **Level 0** → **1** AI engineer.

GAN, GPT, Grounding, Guardrails and more. **Plain-English** definitions, one swipe each, plus **5 G-tools** and **6 G-minds** worth following.

GENERATIVE ADVERSARIAL NETWORK • FOUNDATIONS

GAN

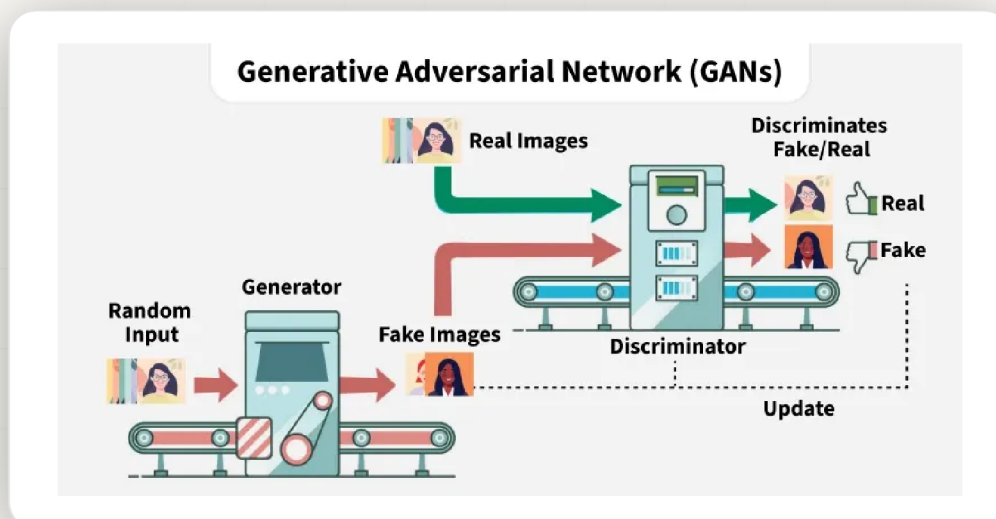
generator • vs • discriminator

Two networks in a fight. One invents, one judges. The generator keeps creating fakes; the discriminator keeps calling them out. They train each other until the fakes are **convincing enough to fool even the discriminator**. That arms race powered the first wave of AI-generated images and deepfakes.

- Image generation

- Synthetic data

- Style transfer



GENERATE • DISCRIMINATE • UPDATE

GENERATIVE PRE-TRAINED TRANSFORMER • ARCHITECTURE

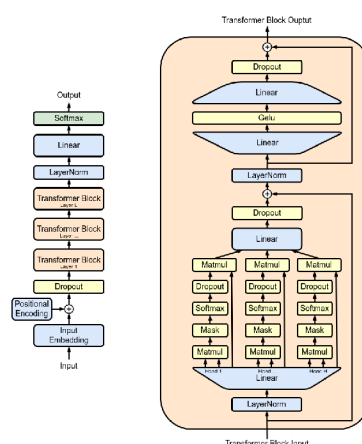
GPT

predict • the • next token

One job: predict the next word. Do that across hundreds of billions of tokens and something remarkable happens. The model starts writing, reasoning, and coding. Fine-tuning then **teaches it to take instructions** rather than just complete sentences. That sequence is how ChatGPT happened.

USED IN

- ChatGPT
- Copilots
- LLM apps



TRANSFORMER STACK

GROUNDING • RETRIEVAL

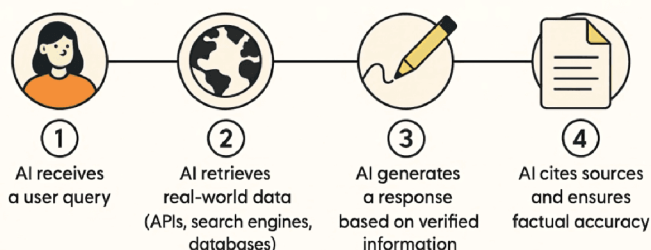
Grounding

retrieve • then • answer

Without grounding, a model answers from memory alone, and memory hallucinates. Grounding **tethers the answer to real sources:** documents, databases, APIs, live search. It is the core idea behind RAG and the primary reason production AI can stay accurate at all.

- RAG systems
- AI agents
- Enterprise search

How Grounding Works in AI



ANCHOR IN REAL DATA

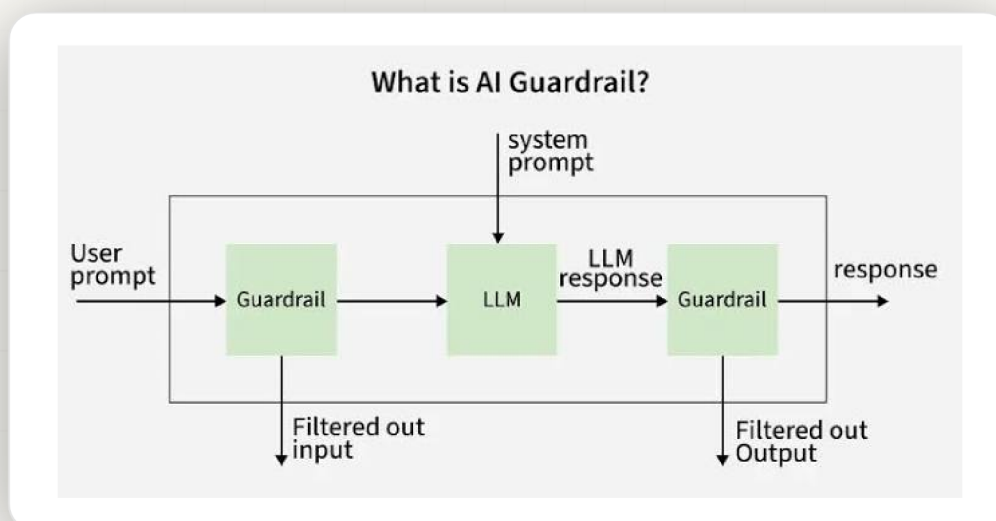
Guardrails

check • filter • constrain

Rules between AI reasoning and real consequences. Guardrails **validate inputs, check outputs, and limit what an agent can touch**. The faster AI moves, the more weight these carry. They are the difference between a useful agent and a liability.

USED IN

- AI safety
- LLMOps
- Agent controls



INPUT → LLM → OUTPUT, CHECKED

GOLDEN DATASET • EVALUATION

Golden Dataset

trusted • ground • truth

Want to know if an AI is actually right? You need answers you already trust. A golden dataset is exactly that: **verified examples with known correct outputs** that every model gets measured against. No golden set, no real benchmark.

USED IN

- Benchmarking
- Regression tests
- Eval sets

INPUT	EXPECTED	✓
"2 + 2 = ?"	4	✓
"Capital of France"	Paris	✓
"Sentiment?"	Positive	✓

VERIFIED GROUND TRUTH

GOSSIP PROTOCOL • INFRASTRUCTURE

Gossip Protocol

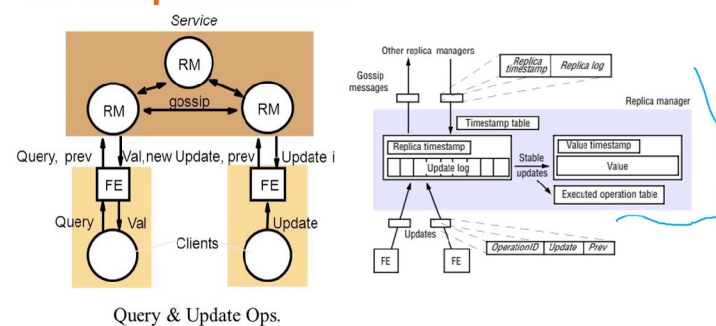
spread • state • peer to peer

Picture how a rumor travels. You tell three people; they each tell three more. A gossip protocol works exactly like that: **each node passes updates to a handful of random peers** until the whole cluster has the news. No central coordinator, no single point of failure.

USED IN

- Distributed systems
- Cluster sync
- Cassandra

The Gossip Architecture



PEERS SYNC BY GOSSIP

↓ UP NEXT • TOOLS

Tools next.

Beyond the **6 G-words**, five G-named tools
shaping how the world builds and ships with AI.

G

MULTIMODAL AI MODEL FAMILY

Gemini

text • image • audio • video • code



Google DeepMind's flagship, built **multimodal from the ground up** and wired into Search, Workspace, and Android, one of the most deployed AI systems alongside GPT and Claude.

WHAT IT IS

Google's flagship multimodal model family.

IN AI STACKS

Chatbots, agents, coding, data analysis.

WHY DEVS USE IT

Long context, strong reasoning, tool calling.

PAIRS WITH

Vertex AI, AI Studio, the Google stack.

REAL-TIME AI ASSISTANT

Grok

by xAI • live web & X data



xAI's assistant, known for a **direct, witty personality and real-time access to the web and X**, strong on current events, reasoning, coding, and image generation.

WHAT IT IS

xAI's conversational AI assistant.

KNOWN FOR

Current events, reasoning, image gen.

WHY PEOPLE USE IT

Real-time answers from the live web & X.

PAIRS WITH

The X platform & xAI's API.

OPEN-WEIGHT MODEL FAMILY

Gemma

by Google • run it locally



Google's **open-weight models** from the same research as **Gemini**, lightweight enough to run on your own hardware, fine-tune, and deploy at the edge.

WHAT IT IS

Open-weight models from Google DeepMind.

FAMILY

CodeGemma, PaliGemma, ShieldGemma.

WHY DEVS USE IT

Local, customizable, strong for its size.

PAIRS WITH

Hugging Face, Ollama, your own GPUs.

AGENTIC AI SEARCH

Genspark

research • synthesize • act



Genspark

An AI-native search platform where [multiple agents](#) gather, reason across sources, and [complete tasks](#), not just return links. A flagship of the agentic-AI movement.

WHAT IT IS

Agentic AI search & deep research.

KNOWN FOR

Multi-agent collaboration, report gen.

WHY PEOPLE USE IT

Structured answers & multi-step tasks.

COMPARES TO

Perplexity, ChatGPT Deep Research.

AI PRESENTATION MAKER

Gamma

The logo for Gamma, featuring the word "GAMMA" in a bold, blue, sans-serif font, centered within a white rounded rectangle.

prompt → deck, doc, or site

An **AI-first PowerPoint alternative** that turns a prompt or document into a polished deck, doc, or website in minutes, content, design, and layout generated for you.

WHAT IT IS

AI presentation, doc & site builder.

KNOWN FOR

Card-based, one-click AI design.

WHY PEOPLE USE IT

Ideas to polished slides in minutes.

BEST FOR

Pitch decks, reports, fast drafts.

↓ UP NEXT • PEOPLE TO FOLLOW

Spotlight next.

Six **G-named minds**, from the godfather of deep learning to the educators shaping how AI gets built and used.

G

HIGH-SIGNAL, LOW-NOISE

Geoffrey Hinton

"Godfather of AI" • deep learning



The mind behind **backpropagation and deep neural networks**. Nobel laureate and Turing Award winner whose work is the foundation under GPT, Claude, and Gemini.

KNOWN FOR

Deep learning, backprop, neural nets.

FIND THEM

cs.toronto.edu/~hinton, talks.

WHY FOLLOW

The founder of modern neural networks.

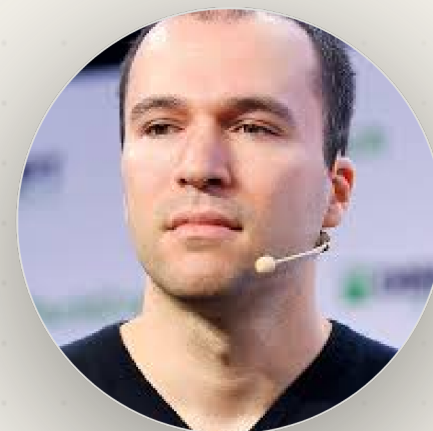
BEST FOR

First-principles intuition on deep nets.

HIGH-SIGNAL, LOW-NOISE

Greg Brockman

co-founder & president • OpenAI



Co-founder and president of OpenAI, the engineer who **helped build ChatGPT, GPT-4, and Codex** and scaled OpenAI from a small lab into one of AI's most influential companies.

KNOWN FOR

OpenAI, ChatGPT, GPT-4, Codex.

FIND THEM

X @gdb, LinkedIn.

WHY FOLLOW

Building at the AI frontier, hands-on.

BEST FOR

Engineering & scaling frontier AI.

HIGH-SIGNAL, LOW-NOISE

Gaurav Sen

system design • AI infrastructure



A tech educator who breaks down **system design, architectural failures, and case studies**, with a focus on scaling safe, sustainable AI infrastructure.

WHAT THEY POST

System design & scaling deep-dives.

FIND THEM

YouTube @gkcs, InterviewReady.

WHY FOLLOW

Complex architecture, made clear.

BEST FOR

Designing scalable AI systems.

HIGH-SIGNAL, LOW-NOISE

Garry Tan

president & CEO • Y Combinator



President & CEO of Y Combinator, a **front-row voice on the AI startup wave**, what early founders are building, funding, and shipping next.

KNOWN FOR

Y Combinator, early-stage AI startups.

FIND THEM

X @garrytan, YouTube.

WHY FOLLOW

Where AI founders are headed next.

BEST FOR

AI startup & founder signal.

HIGH-SIGNAL, LOW-NOISE

Greg Isenberg

AI startups • agents • automation



A creator-founder teaching how to **launch startups with AI agents, no-code, and automation**, building products and one-person companies faster than ever.

WHAT THEY POST

AI business models & agent workflows.

FIND THEM

X @gregisenberg, Startup Ideas pod.

WHY FOLLOW

Turning AI into shipped products.

BEST FOR

AI-native startup ideas.

HIGH-SIGNAL, LOW-NOISE

Ghita El Haitmy

founder • TechBible.ai & Eli



Founder of TechBible.ai and Eli, helping professionals **discover and actually use new AI tools**, one of the most-followed voices in AI-tool discovery.

WHAT THEY POST

New AI tools & how to use them.

FIND THEM

LinkedIn, Instagram @techbible.ai.

WHY FOLLOW

Stay current on the AI-tool landscape.

BEST FOR

Discovering useful AI tools.

The **G** stack, in one card.

Save it for later. Drop a comment if there's a G-term you'd add.

01 **GAN** foundations

02 **GPT** architecture

03 **Grounding** retrieval

04 **Guardrails** alignment

05 **Golden Dataset** evaluation

06 **Gossip Protocol** infra

+ **Gemini • Grok • Gemma • Genspark • Gamma** tools

@ **Hinton • Brockman • Sen • Tan • Isenberg • El Haitmy** minds



FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

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H Next